

Roots, rhizosphere and soil: the route to a better understanding of soil science?

P. J. GREGORY

Scottish Crop Research Institute, Invergowrie, Dundee DD2 5DA, UK

Summary

The centenary of Hiltner's recognition of a rhizosphere effect is a convenient point to assess the impact of such thinking on the direction of soil science. A review of the major soil journals suggests that for much of the last century, Hiltner's insight had little effect on mainstream thinking outside of soil microbiology, but this situation is changing rapidly as the consequences of spatial and temporal heterogeneity on soil functioning assumes greater importance. Studies of root growth, root distributions and of rhizosphere processes over the last 25 years demonstrate both the size and distribution of root systems and the associated inputs from roots to soils. These inputs result in a plethora of dynamic reactions at the root–soil interface whose consequences are felt at a range of temporal and spatial scales. Root growth and respiration, rhizodeposition, and uptake of water and nutrients result in biological, chemical and physical changes in soils over variable distances from the root surface so that the rhizosphere has different dimensions depending on the process considered. At the root length densities common for many crop species, much of the upper 0.1 m of soil might be influenced by root activity for mobile nutrients, water and root-emitted volatile compounds for a substantial proportion of the growing season. This brief review concludes that roots are an essential component of soil biology and of soil science.

Introduction

Roots are almost ubiquitously present in soils, yet they are routinely ignored by many soil scientists, perhaps because while they exist in soil, to many soil scientists they are not of it. The centenary of Hiltner's recognition of the rhizosphere provides a convenient point to assess both the influence of his thinking on the development of soil science, and the gaps in our knowledge of soil–root interactions.

From field and pot studies of the effects of green manuring with legumes on soil fertility, Hiltner (1904) deduced that, to explain his observations, there must be a series of processes occurring at the root–soil interface. Among the processes occurring, it was clear that:

- 1 A volume of soil existed that was shared by roots and bacteria.
- 2 Exuded materials from the roots of different legumes attracted different organisms than roots of non-legumes.
- 3 Each legume species attracted organisms that had a specific benefit for that species.

Hiltner's main insight from these observations was that there was a volume of soil, the rhizosphere, over which the roots had influence and that this soil volume was also shared by bacteria. Apart from a few soil microbiologists, though, soil

scientists in general took some time to incorporate this concept into their mainstream thinking. The role of roots in determining plant accessibility to water and nutrients received little attention throughout the early 1900s when availability of nutrients to plants was defined by use of chemical extractants and that of water was similarly defined by the equilibrium concepts of field capacity and permanent wilting point. It was not until the mid-1950s that ideas of water and nutrient mobility superseded those of equilibria and thermodynamics. For nutrients, the change of thinking came in a groundbreaking paper by Bray (1954) in which he introduced the concept that nutrient mobility was central to soil–plant relations, and demonstrated that mobile nutrients such as nitrate moved to roots from large distances whereas adsorbed nutrients such as phosphate moved only short distances. The corollary of this was that the zones of competition for nutrients by roots differed depending upon the mobility of the nutrient. This change of thinking about the availability of nutrients to plants was paralleled by similar developments regarding the movement of water towards roots (Gardner, 1960).

The consequences of these changes in thinking are evident from the subject matter of scientific papers in the leading soil science journals. Besides a few contributions to mainly biological journals, neither roots nor the rhizosphere impinged on the soil science literature as topics of major importance until the latter part of the 1900s. A quick survey of the major soil

Correspondence: P. J. Gregory. E-mail: Peter.Gregory@scri.ac.uk

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science journals at 5-year intervals from 1950 onwards shows only a very small number of papers about either roots or root systems, or the rhizosphere, in their title or keywords until about 1990 (Table 1). Articles about roots and their activities started to appear in the *Soil Science Society of America Journal* from 1960 onwards (typically one or two per year) but papers on the rhizosphere are still rare (typically less than one per year). As expected, numbers in *Soil Biology and Biochemistry* are higher (typically one per year for roots and seven to eight per year for rhizosphere), with the latter topic appearing as a major area of research from about 1990 onwards. Both topics appear with low frequency in the *European Journal of Soil Science* (less than once per year until 1995 onwards) and *Soil Science. Plant and Soil* has emerged as the journal carrying the most papers on the plant–soil interface, with substantial coverage also in *New Phytologist*.

This paper briefly reviews the current state of knowledge on roots and root–soil interactions, with emphasis on chemical and physical changes in addition to microbiology. It argues that recent understanding of root-induced processes in soil requires greater acknowledgement of roots as an essential component of soil biology if soil science is to advance.

Roots and the soil

Root mass

Root growth is affected by a wide range of soil properties but, in turn, the properties of soils are modified by roots. Over the last 40 years, substantial amounts of data have been collected on the growth of root systems in soils, ranging from the drawings of crop and native plants grown on deep soils by Kutschera (1960), through studies of root development (Klepper *et al.*, 1984), to detailed studies of the seasonal changes in root growth and decay (Gregory & Eastham, 1996; Hendrick & Pregitzer, 1996; Watson *et al.* 2000). There

is a preponderance of studies on cereals among crop plants, and data on the dynamics of perennial plants are comparatively rare. The quantities of root mass and length, and the depth of rooting in soils vary with plant and soil type. A global analysis of root measurements found that average root mass ranged from about 0.2 kg m⁻² for croplands to about 5 kg m⁻² for forests and sclerophyllous shrubs and trees (Jackson *et al.*, 1996). Root mass in forest ecosystems ranged from 2 to 5 kg m⁻², while that in croplands, deserts, tundra and grasslands was < 1.5 kg m⁻². These figures belie the fact that there is a constant turnover of roots and root material, with exudates and other rhizodeposits contributing sustenance to the soil microbial and faunal populations.

Rooting depth

The depth to which roots are able to grow has many implications for the use of water and biogeochemical cycling in ecosystems, and is affected by both genetic and environmental factors. Canadell *et al.* (1996) found substantial variation in maximum rooting depth across the globe, varying from 0.3 m for some tundra and taiga species to 68 m for *Boscia albitrunca* in the central Kalahari. Twenty-two species had roots that extended to ≥ 10 m, with 194 species having roots that were at least 2 m deep. The average maximum rooting depth for each ecosystem ranged from 0.5 ± 0.1 m for tundra to 15.0 ± 5.4 m for tropical grassland and savanna, with deep root habits quite common in woody and herbaceous species across most terrestrial ecosystems. When the plant species were grouped across ecosystems, leaving out the cropped areas, the average maximum rooting depth was 7.0 ± 1.2 m for trees, 5.1 ± 0.8 m for shrubs, and 2.6 ± 0.1 m for herbaceous plants. These results emphasise the vertical extent of root systems in deep soils and emphasize that roots are not confined to the upper horizons of a soil profile.

Table 1 The number of papers dealing with either roots or root systems, or the rhizosphere, as their subject matter in the major soil science journals in 5-year intervals since 1950

period	<i>Soil Science</i>		<i>Soil Biology & Biochemistry</i>		<i>Soil Science Society of America Journal</i>		<i>European Journal of Soil Science</i>	
	Roots	Rhizosphere	Roots	Rhizosphere	Roots	Rhizosphere	Roots	Rhizosphere
1950–54	4	1			1	1	0	0
1955–59	0	0			1	1	0	0
1960–64	0	0			5	1	0	0
1965–69	7	0			5	0	0	1
1970–74	3	0	2	1	6	2	0	1
1975–79	3	1	2	17	5	1	1	0
1980–84	4	1	3	5	8	4	2	0
1985–89	4	0	2	23	6	2	2	0
1990–94	5	2	5	143	11	5	3	3
1995–99	1	0	14	36	11	7	4	6
2000–04	2	2	4	37	9	1	5	6

Root growth and rooting depth of crop plants can be restricted because of physical and chemical impediments (Gregory, 2006). Almost all roots growing through soil experience some degree of mechanical impedance, and if continuous pores of appropriate size do not already exist then the root tip region must exert sufficient force to deform the soil. Roots are often larger than the water-filled pores at field capacity (i.e. pores with diameter $< 60 \mu\text{m}$), so that freely draining pores are the main spaces in which roots can grow. If the soil will not deform, then roots have a limited capacity to modify their anatomy to fit into a pore that is normally smaller than their diameter. In dense soils, then, roots are frequently confined to cracks and structural features and because soil bulk density often increases with depth, roots beneath the cultivated topsoil are frequently observed to cluster in cracks and fissures (Passioura, 1991; Stewart *et al.*, 1999). In addition to the packing density of soil particles, soil strength is also affected by soil water content so that the depth of rain penetration influences rooting depth. For example, in parts of the Mediterranean region, the depth of rooting is frequently determined by the depth of profile re-wetting by rainfall and varies with both site and season so that on different soils in northern Syria, the depths of rooting and of water extraction by barley and chickpea crops was similar at 1.2 m (Gregory & Brown, 1989). Chemical properties such as low pH, large aluminium concentration and salinity also constrain rooting depth where present, though appropriate amendments can sometimes alleviate the problem. For example, the effects of soil acidity in the Oxisols of the Brazilian cerrado were overcome by the addition of Ca as either lime or gypsum, with rooting depth of maize increased from 0.45 to 1.2 m (Ritchey *et al.*, 1980).

Where physical and chemical soil constraints are absent, the maximum depth of rooting on deep soils is genetically determined and differs not only between vegetation types but also between crop species grown under identical conditions. For example, Greenwood *et al.* (1982) grew a range of vegetables on a sandy loam at Wellesbourne, UK, and found that while onion and lettuce roots were confined to the upper 0.65 m, pea rooted to 0.75 m, fababean to 0.85 m, and turnip, parsnip and cauliflower to > 0.85 m. Similarly, Merrill *et al.* (2002) found considerable differences in rooting depth and total root length between eight crops grown at different sites (predominantly a silt loam) over three seasons in North Dakota, USA. Average maximum rooting depth was 1.6 m in safflower, 1.45 m in sunflower, 1.3 m in spring wheat, c. 1.15 m in crambe (*Crambe abyssinica*) and canola (*Brassica rapa*), and 1.0 m in common bean, soyabean and pea.

Root distribution

It is well recognized that roots are not distributed evenly through the soil profile and that their distribution is important in determining the availability of water and nutrients to plants. Jackson *et al.* (1996) found that a simple asymptotic function

could be used to define the cumulative proportion of roots. Their analysis showed that tundra, boreal forest and temperate grasslands had the shallowest rooting profiles, with 80–90% of root mass in the upper 0.3 m of the profile, while deserts and temperate coniferous forests had the least pronounced profiles, with only 50% of root mass in the upper 0.3 m. There were also marked differences between plant species across ecosystems, with grasses having 44% of their root mass in the upper 0.1 m of soil whereas tropical and temperate trees had only 26% in that zone, and shrubs only 21%. The global average distribution of root mass for all biomes and vegetation types was 30% in the upper 0.1 m, 50% in the upper 0.2 m, and 75% in the upper 0.4 m. Schenk & Jackson (2002) extended this analysis to a larger global data base and found that the soil depth containing 95% of roots increased as latitude decreased from 80° to 30° , but in the tropics there was no clear pattern of variation. Deep rooting depths were associated with water-limited environments, with annual potential evaporation and precipitation together accounting for the largest proportion of the variance in rooting depth globally (12% for the depth of 50% of roots and 16% for the depth containing 95% of roots). Overall, the depth containing 95% of all roots was deeper in sandy soils than in clay or loam soils for five of the six vegetation types in which such comparisons were possible and also deeper in ecosystems with shallow organic horizons compared with deeper organic horizons. Interestingly, differences in soil type had only a small effect on the depth containing 50% of roots, probably because this depth is generally within the nutrient-rich topsoil and therefore comparatively unaffected by properties of the subsoil, coupled with the fact that sampling schemes are not usually fine enough to discern subtle differences in the upper 0.3 m of soil (in which the 50% depth usually falls).

In studies with crops, the quantity of root length in layers within the soil profile is normally expressed in terms of a root length per unit volume of soil (L_v , often with units of cm root cm^{-3} soil), frequently referred to as a root length density. Typical values of L_v in the upper 0.1 m of soil are about 20 cm cm^{-3} in grasses, $5\text{--}10 \text{ cm cm}^{-3}$ in temperate cereal crops, and $1\text{--}2 \text{ cm cm}^{-3}$ in other crops, with corresponding differences in mean inter-root distance ($= 1/\sqrt{\pi L_v}$; Table 2). It has frequently been found that roots are distributed in the soil such that their length and mass decrease exponentially with depth. Gerwitz & Page (1974) first proposed this model after

Table 2 Typical root lengths and inter-root distances measured in the upper 0.1 m of soils

Crop	Root length density /cm root cm^{-3} soil	Inter-root distance /mm
Grass	20	1.3
Temperate cereals	5–10	2.5–1.8
Other crops	1–2	5.6–4.0

reviewing literature on vegetable crops, cereals and grasses, and it has been widely adopted since (e.g. Robertson *et al.*, 1993; Zhuang *et al.*, 2001). Figure 1 shows that the distribution of roots of some crops (cauliflower and winter wheat) is well described by such a relation (see Greenwood *et al.*, 1982 for other examples with vegetable crops), but in others (e.g. rape and sugar beet), while this relation can be found in the surface layers, there is a tendency for values of L_v in deeper soil layers to be almost constant. Whether this is strictly a property of the crop or a result of an interaction between the crop and soil properties remains to be established. Such general relations mean that for a few crops grown on deep, uniform soils, generalized functions of root distribution are possible (e.g. winter wheat, Zuo *et al.*, 2004).

To complement measurements of root systems, models have been developed to summarize current understanding so that formal descriptions of root system size, distribution and architecture are available (see Wu *et al.*, 2005 for a recent summary).

The rhizosphere

When Hiltner (1904) first used the term rhizosphere, it was employed in the specific context of the interaction between various bacteria and legume roots. However, it is now appreciated that the rhizosphere is different to the bulk soil due to a range of biological, biochemical, chemical and physical processes that occur as a consequence of root growth, water and nutrient uptake, respiration and rhizodeposition (Hinsinger *et al.*, 2005). The rhizosphere, then, is a 'zone of soil surrounding the root which is affected by it' (Darrah, 1993) but its size

differs spatially and temporally depending on the factor considered, ranging from a fraction of a millimetre for microbial populations and immobile nutrients, to tens of millimetres for mobile nutrients and water, to several tens of millimetres for volatile compounds and gases released from roots. This means that the interface between the root and the soil is complex, frequently an ill-defined boundary, and heterogeneous in space and time.

Rhizodeposits and signalling molecules

Compounds released from roots into the soil change its chemical and physical properties, and stimulate the growth of various organisms. Rhizodeposits of various exudates, sloughed cells and decaying roots provide an important substrate for the soil microbial community and there is a complex interplay between this community and the quantity and type of compounds released (Kandeler *et al.*, 2002; Marschner & Baumann, 2003). Not only are rhizodeposits rapidly utilized by microbes, but their addition to soils can speed up ('prime') the decomposition of native soil organic matter (Paterson, 2003). For example, Ryan *et al.* (2001) demonstrated that when soluble C is added to the rhizosphere at realistic concentrations, rapid mineralization can occur with a half-life for most sugars and amino and organic acids of 0.5–2 hours. This means that photosynthate produced in a shoot may have a half-life of only about 3–6 hours in the rhizosphere.

Estimates of the quantities of rhizodeposits vary quite markedly between studies although values are typically around 20% of the carbon assimilated by photosynthesis (Nguyen, 2003). Larger values of up to 40–50% have been reported (e.g. Lynch & Whipps, 1990), but such large values usually include root respiratory CO_2 , which is not a substrate for microbes. Most studies have been conducted on annual crop species and almost none on mature perennials. One of the most comprehensive field studies was that of Swinnen *et al.* (1995), who pulse-labelled crops of winter wheat and spring barley at regular intervals and allowed the assimilated ^{14}C to be allocated within the plant-soil system for 21 days. A model was used to allocate the C assimilated to various fractions based on the time-dependent specific activity of the daily fixed assimilates (Swinnen *et al.*, 1994) and on the measured rates of respiration of rhizodeposits (Swinnen, 1994). For crops grown under conventional agricultural practice, 18.2% of net assimilation was allocated below ground in winter wheat and 33.3% in spring barley over the growing season (shoot dry weights at maturity were about 1800 and 1080 g m^{-2} , respectively). Over the season, the quantity of C transferred to the roots was 155–225 g C m^{-2} and total rhizodeposition was 65–99 g C m^{-2} (7–15% of net assimilation), representing twice the quantity of roots left at crop maturity. Such results indicate the substantial energy transfer that occurs from annual crops to soil microbes during a growing season.

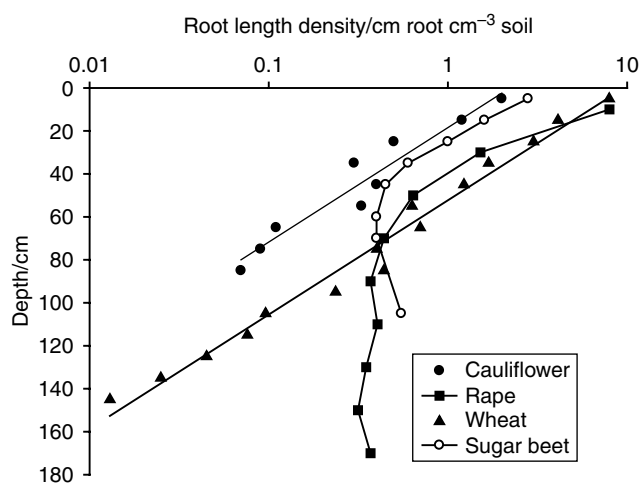


Figure 1 Distribution of root length density with depth in the soil profile for maturing crops of cauliflower (●), oilseed rape (■), winter wheat (▲) and sugar beet (○). Linear regressions have been fitted to data for the distributions of cauliflower and winter wheat roots; oilseed rape and sugar beet data have been joined by straight lines (from Gregory, 2006).

Roots release a wide range of organic compounds, although sugars and polysaccharides, organic and amino acids, peptides and proteins constitute the bulk of the rhizodeposits (Bowen & Rovira, 1999; Jones *et al.* 2004). Much research has now demonstrated that other compounds released from roots may act as messengers that communicate and initiate root-root, root-microbe, and root-faunal interactions (Walker *et al.* 2003). Root-microbe and root-insect interactions can be either positive (symbiotic) to the plant (e.g. via associations with mycorrhizal fungi and N-fixing bacteria) or negative to the plant (e.g. interactions with parasitic plants, pathogenic microbes and herbivorous insects). They involve signal exchange and perception, followed by invasion of the plant by the microbe or insect and concomitant structural changes resulting from the interaction (Mathesius, 2003). Plant roots can also release compounds that are able to disrupt communication between bacteria, thereby reducing their susceptibility to infection (Fray, 2002). For example, successful invasion of roots by a bacterial pathogen often requires a threshold population level of the pathogen to be exceeded. The coordinated bacterial growth that this requires is brought about by a quorum-sensing ability that is achieved by cell-to-cell communication between bacteria using signalling molecules (typically N-acylhomoserine lactones, AHLs, for gram-negative bacteria, and peptide-signalling molecules for gram-positive bacteria). These lactones are found in a wide range of pathogenic, symbiotic and biological control strains of bacteria, with a larger-proportion of AHL-producing bacteria present in the rhizosphere than the bulk soil, suggesting a general role in rhizosphere colonization (Newton & Fray, 2004). Several higher plants have evolved strategies to interfere with this AHL signalling system, including the production of signal mimics, signal blockers, signal-degrading enzymes, or compounds that block the activity of AHL-producing enzymes (Fray, 2002). Mimics of AHL have been detected in a range of higher plants (e.g. Teplitski *et al.*, 2000), suggesting their active interference with bacterial communication.

Microbial changes

Rhizodeposition results in a rhizosphere with different amounts and types of microbial substrates compared with the bulk soil. This, in turn, results in a rhizosphere containing different populations of bacteria, actinomycetes, fungi, protozoa, viruses and nematodes than bulk soil (Lynch, 1990). Foster & Rovira (1976) were among the first to study the spatial relationships between microbial communities and the root by preparing ultrathin sections of wheat rhizospheres. While young roots were only sparsely colonized by microorganisms, by flowering, both the rhizosphere and the outer cortical cells and their cell walls showed considerable development of microorganisms. The numbers, species and sizes of bacteria present in the rhizosphere differed substantially from those in the bulk soil and while the root surface was covered

with individuals, further away the bacteria tended to occur in isolated discrete colonies associated with organic debris.

The composition of the rhizosphere bacterial community is likely to be influenced by the nature of the exudates released (i.e. plant species dependent), location on the root (influencing the quantity and composition of exudates released), and soil type (influencing carbon availability and root-soil adhesion). Marschner *et al.* (2001) investigated the effects of these factors using polymerase chain reaction-denaturing gradient gel electrophoresis (PCR-DGGE) to characterize the bacterial community composition of three plants (chickpea, rape and Sudan grass) grown in three soils (a sand, sandy loam, and clay). Analysis of 16S rDNA band profiles showed that the three plants had similar indices of bacterial diversity but that diversity was significantly greater in the mature root zone than at the root tip in the sandy soil and the clay, but not in the loamy sand. Canonical correspondence analysis showed significant effects of soil type, plant species and root zone on the bacterial community structure; soil and root zone effects cumulatively explained 84.7, 83.0 and 81.6% of the total variance in chickpea, rape and Sudan grass, respectively. In chickpea, the community composition of the sandy soil differed from that of the other two soils, but for the other two plants root zone affected composition more than soil type. In the sand and sandy loam, community composition was influenced more by the root zone than the plant species, whereas in the clay soil the community associated with chickpea differed considerably from that of the other plant species. These complex interactions mean that the relative importance of any single factor, for example soil type, on bacterial community structure cannot yet be specified. Similar PCR-DGGE analysis has also demonstrated changes in bacterial composition and functional diversity across the rhizosphere (Kandeler *et al.*, 2002). Communities in unplanted controls were unaffected by distance, but with plants there was a gradient in composition within the first 2.2 mm from the root surface. The composition was also different from that further away from the root.

Chemical changes

Chemical changes in the rhizosphere of plants have been widely reported (see Hinsinger, 1998, 2001, for reviews), though their quantitative significance for crops is still a matter of debate. For plants grown on soils deficient in P, a wide range of chemical mechanisms exists to increase P availability, including the release of organic anions. For example, upland rice, which is often grown on acidic, P-deficient soils was found to take up P from alkali-soluble inorganic forms when grown on a highly weathered, P-deficient soil, possibly as a result of organic anion release (Hedley *et al.*, 1994), and also to release citrate, together with smaller quantities of oxalate, malate, lactate and fumarate, when grown in thin layers of the same soil supplied with P-free nutrient solution (Kirk *et al.*, 1999). Apparent rates of citrate release ranged from 337 to

155 nmol g⁻¹ root fresh weight hour⁻¹ over the course of plant growth equivalent to 2–3% of plant dry weight. Kirk *et al.* (1999) demonstrated that the measured P uptake by the plants could be quantitatively accounted for by P solubilization by citrate, and that the main mechanism of solubilization involved either the chelation of metal ions that would otherwise have immobilized P or the formation of soluble citrate-metal-P complexes or both. Plants that form proteoid or cluster roots are of particular interest because they release a range of organic anions (Lamont, 2003). While relatively unimportant among crop plants, except in some species of the genus *Lupinus*, cluster root formation is common among slow-growing sclerophyllous shrubs and trees in western Australia and South Africa, which are adapted to habitats of very small chemical fertility and with P as a major limiting nutrient (Pate & Watt, 2002).

Physical changes

Physical changes in the rhizosphere have been much less studied than biological or chemical changes, despite their potential consequences for the movement of water and solutes (Gregory & Hinsinger, 1999). The biological and physical interactions occurring in the rhizosphere, especially as a consequence of root growth, rhizodeposition, microbial activity, and the repeated wetting and drying of soil at the root-soil interface, result in a heterogeneous soil matrix with physical properties that are different to soil at some distance from the root. For example, structural differences between bulk and rhizosphere soil have been reported with the rhizosphere containing more larger pores (Whalley *et al.*, 2005), and in a silty soil, soil adhering to maize roots had greater aggregate strength (450–500 kPa) than that not adhering (410–420 kPa; Czarnes *et al.*, 2000a). The formation of rhizosheaths around some roots (especially grasses) is a widely recognized structural feature induced by roots and relies on immature xylem vessels, root hairs, and the release of water and mucilage from roots (Figure 2; McCully, 1999). Mycorrhizal hyphae are also implicated in the adhesion of soil particles to roots, together with the exopolysaccharides produced by rhizosphere microorganisms (Amellal *et al.*, 1998).

When a root grows, it deforms soil by expanding radially, and the volume occupied by the root is matched by an equivalent loss of pore space from the surrounding soil. Dexter (1987) predicted an exponential decrease in density with distance from the root-soil interface within a homogeneous soil matrix. Bruand *et al.* (1996) used backscattered electron scattering images (resolution 60 µm) of maize roots growing in remoulded silty clay loam and clay loam soils to quantify changes in porosity and found that porosity was 22–24% less within the soil adjacent to the root than in the surrounding soil. Bulk density increased to 1.80 Mg m⁻³ at the root-soil interface compared with 1.54 Mg m⁻³ at 1 mm from the root. The limited distance over which changes occurred in this study

(about 800 µm) contrasts with the effects over 4–5 mm summarized by Young (1998) and with the predictions of Dexter (1987), and is probably a consequence of using remoulded materials. More studies of naturally structured soils at fine resolution are required to resolve both the size of changes that occur and their potential consequences for liquid and gaseous flow at the root-soil interface.

The release of root mucilage may also change the water relations of the rhizosphere. Passioura (1988) suggested that this might be particularly beneficial for water uptake at small values of matric potential (dry soils). Read & Gregory (1997) showed that mucilage reduced the surface tension of water and Read *et al.* (2003) confirmed that the addition of the surfactant component of mucilage can alter the relationship between water content and soil matric potential (the moisture characteristic curve), making the soil drier at a given value of matric potential, especially at high matric potentials. Whalley *et al.* (2005) also found that rhizosphere soil of maize and barley tended to be drier at a given matric potential than bulk soil but suggested that differences in wetting angle and pore connectivity were the likely explanation for these differences. There have been very few direct studies of rhizosphere hydraulic properties *in situ*, although Hallett *et al.* (2003) used a miniaturized infiltrometer to study the hydraulic characteristics of barley, rape, potato and grass rhizospheres. In excavated blocks of



Figure 2 Cryo-scanning electron micrograph of the rhizosheath surrounding a buckwheat (*Fagopyrum esculentum*) root growing in the field. Root hairs can be seen crossing a macropore. I am grateful to Dr M. McCully for this previously unpublished figure.

field soil there were significant differences between plant species in sorptivity and water repellency, but no differences between soil within 1 mm of the root surface (rhizosphere soil) and that 20 mm from a root were detected. The challenge of assessing the importance of rhizosphere-scale changes to whole root systems is being taken up through the development of models (e.g. Kirk *et al.*, 1999, for chemical changes; Dunbabin *et al.* 2005, for physical changes; Darrah *et al.* 2006, for general discussion), but more research is required to resolve the size and direction of any changes in rhizosphere structural and hydraulic properties and to quantify their significance for field-grown plants.

Root-induced soil processes

Vegetation is considered in the pedology literature to be one of the six major factors giving rise to different types of soil (Jenny, 1941). Many of these long-term effects are a consequence of the different properties of leaf and shoot components rather than roots *per se*, but over extended periods, roots have a major influence on the formation of soils. Pedogenesis results in heterogeneous distributions of soil materials and nutrients leading to potentially important consequences for plant roots. For example, Qureshi & Jenkins (1987) found that P and S were concentrated at ped surfaces in two Inceptisols and an Alfisol so that roots exploiting the voids between peds will be exposed to greater nutrient concentrations than those existing in the ped interiors.

Mineralogical transformations

Chemical weathering of minerals to form soil materials may also be enhanced by the presence of roots and their associated microflora, although, because of the time scales involved, field-based evidence of such processes is rare. Root-induced vermiculitization of the mica phlogopite was demonstrated under laboratory conditions by Hinsinger & Jaillard (1993), and weathering of vermiculite has also been demonstrated by cultures of ectomycorrhizal fungi (Paris *et al.*, 1995). A range of processes may be involved, including the release by roots of protons and organic anions, and the depletion of cations to concentrations small enough to destabilize crystal lattices. In the laboratory study of potassium release from phlogopite by ryegrass, the equilibrium concentration of K in the soil solution below which the mica became unstable was about 80 $\mu\text{mol l}^{-1}$, but smaller concentrations would be required if the dominant micas were of the more commonly occurring dioctahedral soil minerals muscovite (equilibrium concentration 2–5 $\mu\text{mol l}^{-1}$) and illite (25 $\mu\text{mol l}^{-1}$; Hinsinger & Jaillard, 1993). In rape, by contrast, irreversible transformation of phlogopite to vermiculite was brought about by severe root-induced acidification of the rhizosphere, leading to acid dissolution of the phlogopite lattice (Hinsinger *et al.*, 1993). Plant roots, then, may be responsible for specific forms of weathering that are

different to those occurring in the bulk of the soil and may explain why the clay mineralogy of the rhizosphere is sometimes reported to be different from that of the bulk soil (April & Keller, 1990).

Aggregation

The development of water-stable aggregates is an important process in the genesis of soils because it strongly influences a range of soil characteristics, including aeration, infiltration and erodability. Plant roots play a major role in this process. Their influence comes about indirectly through the release of carbon compounds that provide a substrate for microbes (Young & Crawford, 2004), and directly through: (i) wetting and drying phenomena; (ii) the accumulation in some soils of inorganic chemicals at the root surface that act as cementing agents; (iii) the release of organic compounds that promote aggregation of particles; and (iv) the structural support of undecayed, senescent roots that act like steel rods in reinforced concrete. Tisdall & Oades (1982) showed that the water stability of aggregates in many soils was dependent on organic materials with roots and fungal hyphae (i.e. growing root systems) important in the stability of macroaggregates (> 250 μm diameter). The numbers of stable macroaggregates decreased with organic matter content as roots and hyphae decomposed, and were related to management practices, with increases under pasture and decreases under arable cropping. In contrast, the stability of microaggregates was determined by the content of persistent organo-mineral complexes and by more transient polysaccharides, leading to their relative insensitivity to changes in soil organic matter content caused by different management practices.

Cycles of soil wetting by rain and drying by plant roots also have a big effect on aggregation (Horn & Smucker, 2005). Materechera *et al.* (1992) found that aggregation in two soils (a Luvisol and a Vertisol), initially dried and sieved to 0.5 mm, was influenced by soil type, plant species and wetting and drying cycles in a controlled experiment over a 5-month period. Denser and more stable aggregates were formed in the Vertisol, but for both soil types wetting and drying cycles and greater root length increased the proportions of smaller aggregates and aggregate strength compared with unplanted soil. Root length was in the order ryegrass > wheat > pea, which was also the order of water stable aggregates > 0.25 mm in diameter (Table 3). Drying results in enhanced adhesion by capillary forces, leading to greater cohesion as mineral and organic materials are drawn into closer contact. For example, Czarnes *et al.* (2000b) examined the interaction of exudates and wetting and drying using two model bacterial exopolysaccharides (dextran and xanthan) and a root mucilage analogue (polygalacturonic acid) mixed with soil dried and sieved to <2 mm diameter. Xanthan and polygalacturonic acid increased the tensile strength of the soil over several wetting and drying cycles, suggesting that they increased the bond

Table 3 The influence of plant species and water regime on the stability of aggregates. The results are expressed as the proportion of the > 0.25 mm fraction stable to wet sieving (from Materechera *et al.*, 1992)

Plant species	Luvisol			Vertisol		
	Continuously wet	Wetting and drying	Mean	Continuously wet	Wetting and drying	Mean
Control	0.245	0.314	0.280	0.561	0.674	0.617
Pea	0.417	0.495	0.456	0.732	0.876	0.804
Wheat	0.468	0.556	0.512	0.793	0.853	0.823
Ryegrass	0.597	0.644	0.620	0.829	0.909	0.869
Mean	0.432	0.502		0.729	0.822	
LSD, $P = 0.05$						
Water regime		0.013			0.021	
Plant species		0.009			0.015	

energy between particles. There were differences between the mucilage analogues in their modes of action, so that while wetting and drying increased sorptivity and decreased repellency with dextran and xanthan, polygalacturonic acid reduced the rate of wetting. Overall, polygalacturonic acid appeared to stabilize rhizosphere soil structure by simultaneously increasing the strength of bonds between particles and decreasing the wetting rate. While the effects of wetting and drying on the types and properties of structures produced matched qualitatively with field observations, some caution is required in extrapolating these results to field conditions because polygalacturonic acid does not replicate exactly the behaviour of root mucilage, and microbial degradation of polysaccharides released by roots and microbes may restrict their persistence in soils.

Turnover of soil organic matter

Microbial communities in soils are a key influence in regulating the dynamics of organic matter decomposition and the availability of plant nutrients such as nitrogen, phosphorus and sulphur (Paterson, 2003). As described earlier, the release of rhizodeposits changes the microbial community in the rhizosphere and this, in turn, affects the rate of turnover of soil organic matter and the release of nutrients compared with the bulk soil. Different types of rhizodeposit are released from different parts of the root system, with recent assimilates lost mainly as exudates from near the root tip while more complex materials are lost from more mature parts of the root system in cell sloughage and root turnover (Jaeger *et al.*, 1999; Thornton *et al.*, 2004). This change of substrate can have a large effect on the microbial community and on its activity. For example, Clayton *et al.* (2005) found that PCR-DGGE profiles of bacterial and fungal communities were different for regions near the root tip and those of mature roots for both perennial ryegrass and white clover. Tracking the specific microbial groups responsible for the utilization of particular compounds has not been possible until recently, but the development of

RNA-stable isotope probing offers a possible way forward. Rangel-Castro *et al.* (2005) found that the technique was able to detect differences between microbial communities associated with upland grassland grown on either limed or unlimed soil. On the limed soil, the majority of the community (bacteria, archaea and fungi) appeared to utilize root exudates, while on the unlimed soil a proportion of the active community was utilizing other sources of organic carbon. This technique may allow investigation of the role of specific microbial groups in carbon turnover within the rhizosphere.

Conclusions

This brief review indicates that there is a plethora of dynamic processes occurring at the root–soil interface, whose consequences are felt at a range of temporal and spatial scales. The centenary of Hiltner's recognition of a rhizosphere provides a convenient point to assess the direction of soil science. Many past classical approaches in soil science, especially in the use of soils for crop production, have served to minimize this dynamism by dealing with equilibrium measurements in homogenized soils (e.g. the use of chemical extractants on < 2 mm sieved soil to assess nutrient availability), but the recognition of spatial gradients and of heterogeneity within soils requires new approaches. The rhizosphere has sometimes been depicted as a soil cylinder of given radius around the root, but drawing a boundary between rhizosphere and bulk soil is impossible because different processes result in gradients of different sizes (Hinsinger *et al.*, 2005). From the knowledge that we now have about root systems growing under field conditions in soils used for managed production systems, the quantities of root present suggest that the distance between roots is small enough to mean that for the purposes of mobile nutrients, water, and volatile compounds, most of the soil in the upper 0.1 m of the profile could be regarded as rhizosphere soil for at least a substantial part of the growing season. The size of the rhizosphere for microbes and immobile nutrients will, of course, not equate to the whole soil mass and further

research is required to determine whether the spatial gradients measured in laboratory experimental systems are also evident in structured soils.

Roots are a source of substrates and energy for many microbes and act as agents for biological, chemical and physical changes in soils; they are an essential component of soil biology and of soil science.

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